

Recap

Let's start by recalling the following algebraic properties,

Algebra properties

$$\begin{aligned}
 a + b &= b + a, & (a + b) + c &= a + (b + c), & a(b + c) &= ab + ac, \\
 \frac{a}{b} + \frac{c}{d} &= \frac{ad + bc}{bd}, & \frac{a}{b} \cdot \frac{c}{d} &= \frac{ac}{bd}, & \frac{-a}{b} &= -\frac{a}{b} = \frac{a}{-b}, & \frac{\frac{a}{b}}{\frac{c}{d}} &= \frac{ad}{bc} \\
 \sqrt{ab} &= \sqrt{a}\sqrt{b}, & \sqrt{\frac{a}{b}} &= \frac{\sqrt{a}}{\sqrt{b}}, & \sqrt[n]{a} &= a^{1/n} \\
 a^2 - b^2 &= (a - b)(a + b)
 \end{aligned}$$

Laws of exponents

$$a^r \cdot a^s = a^{r+s}, \quad (a^r)^s = a^{rs}$$

$$\frac{a^r}{a^s} = a^{r-s}, \quad (ab)^r = a^r b^r$$

$$\left(\frac{a}{b}\right)^r = \frac{a^r}{b^r}, \quad b \neq 0$$

Laws of Logarithms

$$\log_a(AB) = \log_a(A) + \log_a(B)$$

$$\log_a\left(\frac{A}{B}\right) = \log_a(A) - \log_a(B)$$

$$\log_a(A^C) = C \log_a(A)$$

Inverse functions property

$$(f \circ f^{-1})(x) = x = (f^{-1} \circ f)(x) \quad \rightarrow \quad \log_a(a^x) = x = a^{\log_a(x)}$$

Notation

Interval notation. To denote intervals we can use the following notation: $x \in (a, b)$, $x \in [a, b]$ or $a < x < b$. $a \leq x \leq b$. Remember that $(,)$ and $<$ are used for open boundaries, that is that the variable x can not take that value. Meanwhile, $[,]$ and $\leq$ are used for closed boundaries, that is that the variable x can take that value.

Class Activity

$$\text{Simplify: } \frac{x^2}{x-4} - \frac{x+1}{x+2}$$

$$\text{Factor: } x^3y - 4xy$$

$$\text{Solve: } \frac{2x}{x+7} = \frac{2x-7}{x}$$

Function composition $(f \circ g)(x)$;

$$f(x) = x^2 + 3x - 8; g(x) = 3x - 5$$